

Technical Specification: Document Ingestion Pipeline

1. Overview

The Document Ingestion Pipeline (`docs_import.py`) serves as the entry point for the RAG data store. Its primary objective is to process incoming unstructured documents, normalize their text content, and prepare them for downstream vector generation and database storage.

2. Functional Alignment (FR-2)

The module strictly fulfills Functional Requirement 2 (Document Import) by executing the following sub-tasks:

- **Multi-Format Parsing:** Dynamically handles text extraction for `.txt`, `.pdf`, and `.docx` extensions.
- **Streamlined Normalization:** Strips trailing spaces, eliminates layout artifacts, and verifies data integrity.
- **Pipeline Interactivity:** Implements a Command Line Interface (CLI) to accept dynamic file paths at runtime.

3. Core Architecture & Functions

`extract_text(file_path)`

Evaluates file extensions using `splittest` logic and routes execution to the appropriate extraction library (`pypdf` for PDF data streams, `python-docx` for Microsoft Word structures, or native utf-8 stream readers for text files).

`import_document(file_path)`

Orchestrates the ingestion lifecycle: extracts raw string data, triggers the text chunking mechanism, interfaces with the Sentence-Transformer module, establishes a transaction block via `psycopg2`, and executes relational SQL compilation targeting the target datastore.

4. System Dependencies

- `pypdf`: High-fidelity binary PDF stream parsing.
- `python-docx`: Element extraction for structured OpenXML documentation formats.

- `psycopg2-binary`: Thread-safe PostgreSQL connectivity and client transaction execution.